ELEC5305 – How MVDR beamforming performance can be improved compared to conventional beamforming for underwater acoustic signals

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Github Project Link: https://github.com/Jdul9044/ELEC5305-Project-Jdul9044

Project Overview

Currently in Digital Signal Processing (DSP), particularly in audio and acoustics the quality of a signal still remains to be an area of study and for future development. One way in understanding quality is to investigate its Signal to Noise Ratio (SNR). An audio recording with a high level of noise can contaminate the audibility and potentially make adjusting the gain redundant. In an amplifier if there is a low SNR in the input of the system as the gain is increased, the noise is also increased, hence making it hard to remove noise when it is already in the system. For higher grade audio equipment, a lower noise floor is more attractive for studio application, however, for inexpensive products this may not be achievable. Consequently, for more dynamic environment such as underwater the issue with noise becomes more complex with interference caused by multipathing of the noise as well as can be affected by ambience. There are multitude techniques applied into optimising and improving SNR. One way is to “add some spatial information to the time frequency” which is achieved using multiple sensors, this is called beamforming [1]. For the basis of this project the intention is to demonstrate how a conventional beamformer can achieve SNR improvements in a noisy underwater environment and possibly evolving the computation into an adaptive beamformer such as Minimum Variance Distortionless Response (MVDR). For underwater environments adaptive beamforming will have significant benefits as these multipaths could potentially be suppressed and improve the SNR.

Background and Motivation

The use of beamforming was a technique not well understood until the early 1900s by Guglielmo Marconi and Karl Ferdinand Braun and then later in the 1960s became introduced in sonar phased array systems [2]. After the 1960s adaptive beamforming methods were starting to be implemented which essentially are algorithms that adjust based on the environment through statistics and attempt to negate other interference or noise [2]. Although beamforming has mostly been a solved problem, standard conventional beamformers are still used today for their robustness. In addition, there also has been studies made on improving robustness as well on MVDR [2]. Due to the advancements of faster Analogue to Digital Converters (ADC) and easier to obtain electronics, access to digital beamforming has made both types of beamforming widely used today [3]. This DSP technique was chosen for exploration of complexity and for its wide use in audio application especially in underwater acoustics.

Literature Review

Beamforming, although a very powerful technique in improving a SNR of signal, can come with limitations especially for a linear array configuration. One limitation is mentioned in “Beamforming Using Digital Piezoelectric Mems Microphone Array” which details how spatial aliasing can be caused by geometric constraints [4]. Spatial aliasing can be described as a phenomenon where directional information is limited to a particular frequency band and can cause artifacting in the data from grating lobes.

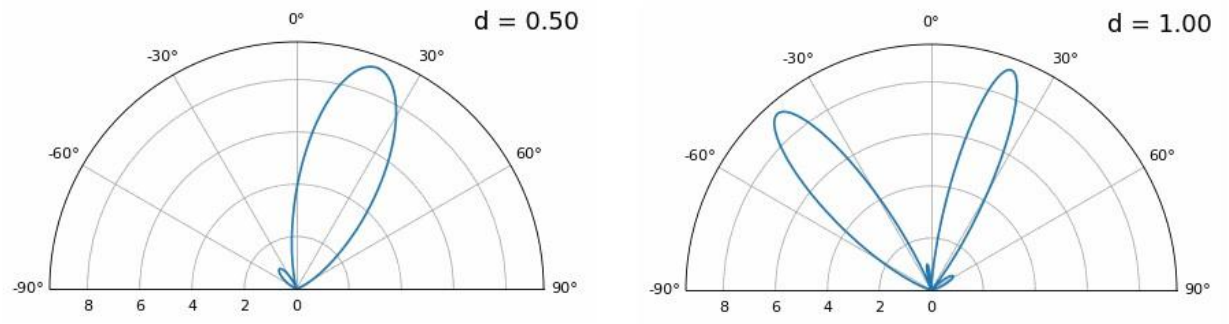


Figure 1. (left) Beamforming without spatial aliasing, (right) grating lobes becoming introduced [5]

Consequently, spatial aliasing can have a profound effect on the data if not well understood. To beamform with an optimum performance and avoid spatial aliasing the paper suggests keeping an array with at least a spacing of less than half a wavelength as shown by the following [6]:

$$d \leq \frac{\lambda}{2}$$

Where d is the element spacing and λ corresponds to the wavelength. As a result, the best theoretical spacing to avoid grating lobes can be found by the following:

$$d \leq \frac{\left(\frac{c}{f_{max}}\right)}{2}$$

Conversely, the beamformer performance on the lower frequency end can be reduced, this is due to the fundamental limitation of angular resolution. The angular resolution determines how the main lobe beamwidth is adjusted. The paper “Simulation and design of a microphone array for beamforming on a moving acoustic source” quotes how the main lobe is widest” at the minimum frequency [6]. Hence, for a narrower main lobe the distance spacing should also not be too low.

Additionally, the paper titled “Angular Resolution Enhancement Provided by Nonuniformly-Spaced Linear Hydrophone Arrays in Ocean Waveguide Remote Sensing” explores how the use of multiple arrays with different spacings into one can achieve performance improvements in the angular resolution [7]. By performing beamforming with different element spacing the main lobe widening can be prevented.

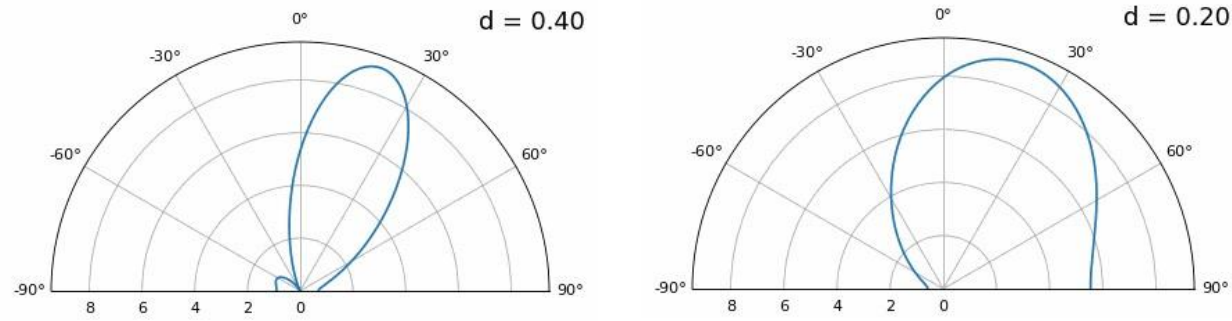


Figure 2. The effect of spacing on angular resolution [5]

Conventional beamformers tend to have the most robust implementation, however, utilise uniform weighting that can potentially include sidelobes that could affect the data. The use of spatial tapering can be applied for the number of elements. Although side lobe reduction is improved, there is a trade-off with the main lobe beamwidth being increased and the gain being reduced [7]. In addition, the paper “Hierarchical Beamforming Strategies for Source Detection With Linear Arrays at 26 GHz” talks about how to achieve high sidelobe suppression performance the need for much larger number of elements is required [8].

For a standard linear array one of the problems beamforming can have is inconsistency in the shape of the main lobe across the frequency. For example, “A Three-Dimensional Pattern-Space Representation for Volumetric Arrays” goes in depth in relation to how the main lobe broadens the most at a steering angle facing end-fire [9]. For volumetric and planar array this is not the case as the main lobe remains consistent regardless of the steering angle [9]. One of the main consequences of the linear array is addressed in the paper “Left-Right Ambiguity Resolution of a Towed Array Sonar” which details how a linear array is limited to a “cone” shape restricting the array to 180 degrees, having a “left” or “right” ambiguity [10].

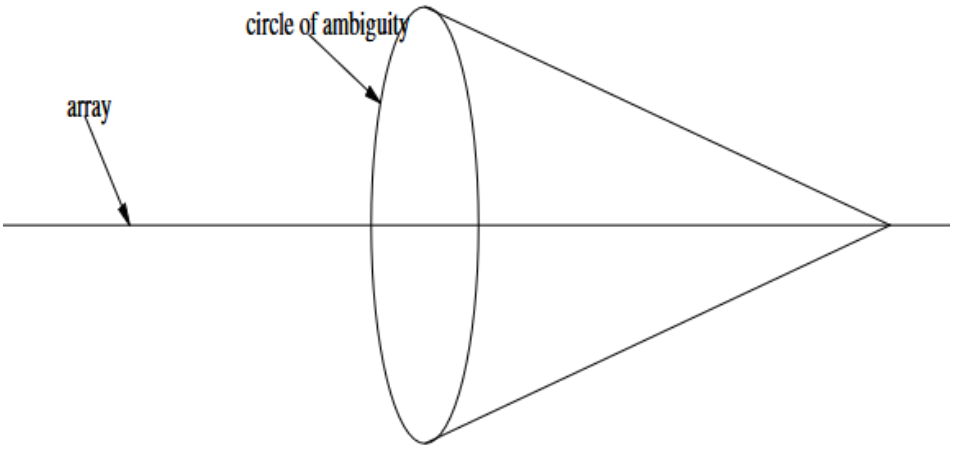


Figure 3. Linear Array Ambiguity [10]

Beamforming with an adaptive beamformer such as MVDR can pose a challenge when faced with errors in measurements such as positional information. “Digital Beamforming with Non-ideal Random Volumetric Arrays” details how the SNR performance of conventional beamformers can potentially be greater than MVDR “in presence of steering vector error” [11]. In addition, having an accurate covariance matrix is another important component to perform MVDR beamforming which is mentioned in “MVDR Algorithm Based on Estimated Diagonal Loading for Beamforming” [12]. This paper addresses on of these fundamental issues by introducing diagonal loading into the covariance matrix to stabilise these problems. Although the use of diagonal matrix can be useful, a small “snapshot” can introduce “large inaccuracy in the sample covariance matrix [12]. Due to the variance of time “snapshots” in the data caused by different number of sensors potentially for different frequency bands, a more adaptable diagonal loading is required [12].

Proposed Methodology

Tools and Platforms

For the main purpose of this project report the analysis will be conducted in Jupyter Labs with Python. Libraries such as Numpy, Scipy and Matplotlib will be used as the main packages. However, generated simulated data will be produced using Python scripts. The main code will be developed for a post-processing environment. ARL Python Tools will be used for validation.

Setup

The current setup of the beamforming array for a simulation will be a fixed spacing for a set of elements. Due to the loss in angular resolution the frequency will also be limited to a low frequency region. For an optimum array for beamforming, the following spacing conditions are required:

$$d \leq \frac{c}{(2f_{max})}$$

$$d \leq \frac{1500}{(2 \cdot 400)}$$

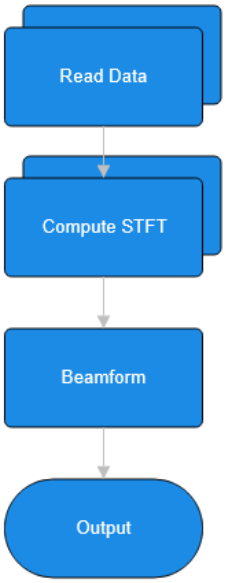
$$d \leq 1.875m$$

The speed of sound in water will be assumed to be at approximately 1500 m/s, with a selected frequency band of 0 to 400 of interest.

For the simplicity of the analysis, a uniform linear array will be tested for the simulation where no angle greater than 180 degrees in azimuth will be selected with a flat 0-degree elevation. Due to the spatial ambiguity constraints of the linear array for real linear arrays especially underwater, the target will not be on the sea floor and be moving above, which will be moving relative to the array axis, thus restricted to an angle of a cone.

DSP Technique

The overall analysis will consist of signal processing chain as followed:



Read Data - The simulated data is initially read as an individual audio Pulse Code Modulation (PCM) file for each channel with a bit-depth of at least 16-bit is selected for minimal quantisation effects [13].

Downsample - Due to the enormous size of the samples used for a long duration with a sampling frequency of 48 kHz dataset, the data can be downsampled into 2000 Hz, given the frequency of interest is fixed to the practical geometric limitation of the array for beamforming. By downsampling the data, the post processing time is effectively reduced and optimised.

Compute Short-Time Fourier Transform (STFT) - The analysis will consist of beamforming in the frequency domain is selected as it is the fastest method for computation for post processing. The data that will be used for the beamforming will have STFT coefficients in the form of complex to utilise the phase information and be converted into Power Spectral Density (PSD) for a proper representation of how each bin the energy is stored in a spectrogram. A discrete-time STFT will be the main method of computing the spectrogram with a standard Hann Window and a 50% overlap choice. Window sizes will be selected for narrowband and wideband for performance metrics.

Beamforming - A conventional beamformer will be implemented for the frequency domain that will be generalised to work with both linear and volumetric arrays. In addition, the conventional beamformer will be compared for an adaptive beamforming using MVDR.

One of the foundational important components for beamforming needed is finding the time delay with respect to each element to a reference. In 3D space, the projection angle of the planar waves can be determined by finding the dot product of the array geometry with the unit vector which is simply the target of interest in spherical coordinates. This is shown below:

$$\tau = \frac{\vec{r} \cdot \vec{u}}{c}$$

Where $\vec{r}$ is the position matrix of the array in the following form:

$$\vec{r} = \begin{bmatrix} x_1 & y_1 & z_1 \\ \vdots & \vdots & \vdots \\ x_n & y_n & z_n \end{bmatrix}$$

Where $\vec{u}$ is the unit vector following an adjusted spherical coordinate system convention as followed.

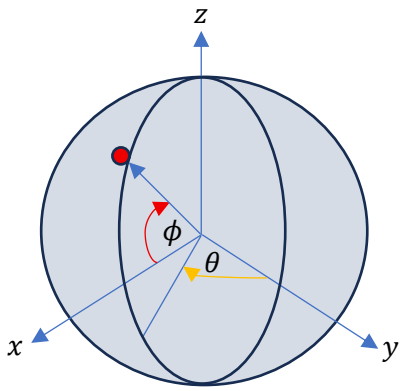


Figure 4. Adjusted spherical coordinate system for underwater [14]

This translates into the following vector:

$$\vec{u} = \begin{bmatrix} \cos\phi \sin\theta \\ \cos\phi \cos\theta \\ \sin\phi \end{bmatrix}$$

Where ϕ is the elevation and θ is the azimuth angles. For the basis of the report, the hydrophone array will be defined by the following convention for reference:

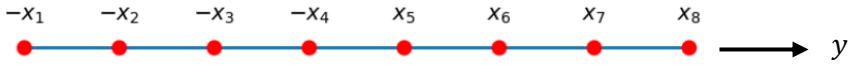


Figure 5. Hydrophone Reference

The hydrophone reference is taken from the centre, where the first half of the array is negative and second half become positive. To compute the steering vector, the phase information is stored as a function of frequency for each element in the vector.

$$s = e^{-j2\pi f\tau}$$

Where s is the steering vector, f is the selected frequency bin and τ being the time delay previously mentioned.

For a conventional beamformer, the weights that are applied to the steering vector. Given the nature of the underwater environments having low SNR, the use of spatial tapering will not be considered for the report. Hence, resulting into beamformer output to the following:

$$Y = conj(s) \cdot X$$

Where the dot product is performed once again to the conjugate of the steering vector with the spectrogram coefficients of each element for a particular time slice.

To implement adaptive beamforming (MVDR) the weights for the steering vector are adapted by the covariance matrix. The “spatial” covariance matrix can be found by the following:

$$R = \frac{XX^H}{N}$$

Where X refers to the spectrogram that is multiplied by itself with the Hermitian matrix divided by N the number of snapshots. Note that for a non-stationary target, using all the time samples from the recording cannot be used for a single covariance matrix. A new method will also be adopted to use multiple time samples which can be done by implementing a moving averaging window. In addition, if the covariance matrix is ill-conditioned as previously discussed, a diagonal loading is applied. For this report, a robust adaptive diagonal loading is applied as followed:

$$R_{loaded} = R + loading \cdot I$$

Where the loading is multiplied by the identity matrix. To compute the MVDR weights the covariance matrix is used into the following equations:

$$w_{mvdr} = \frac{R^{-1}s}{s^H R^{-1}s}$$

Which becomes:

$$Y_{mvdr} = conj(w_{mvdr}) \cdot X$$

Synthetic Production – To test the beamformer, a simulated dataset is produced. To develop the best possible results the simulated dataset will includes tones as well as a broadband inject signals into the files.

A pure tone can be produced by the following:

$$x(t) = \text{Asin}(2\pi f(t - \tau)$$

Where A is the amplitude, f is the frequency and τ is the time delay.

For a broadband signal:

$$x(t) = FFT^{-1}\left(X(f,t)e^{-1j2\pi f\tau}\right)$$

Where $X(f,t)$ is a spectrogram that contains random noise generated between a frequency band.

Validation - To generate a single value error, the absolute difference in error from the spectrum of the validated tools will be subtracted from the main spectrum of the developed code across each frequency then later averaged.

$$mean\ absolute\ error = mean(|spectrum(f) - spectrum_{validated}(f)|)$$

Where the spectrum is time averaged for both the main and validated code.

Superdirective Beamforming – To study the performance metric of the real data of the hydrophone array, an ideal case will be considered for MVDR. To show best results in the capability of the array it will be assumed there are no directive interference and that the field is “perfectly diffuse” hence, behave similar to a superdirective beamformer [15]. The MVDR covariance matrix will be replaced with the diffuse noise covariance to compute the result by the following [15]:

$$R = \frac{\sin\left(\frac{2\pi f}{c}|x_m' - x_m|\right)}{\frac{2\pi f}{c}|x_m' - x_m|}$$

Where f is the frequency, c is the speed of sound, $x_m{}'$ and x_m are the distance of the sensor pairs.

White Noise Gain (WNG) – One of the performance metrics for beamformers that will be compared is the white noise gain. The WNG is calculated based on the weighting applied to the steering vector as shown:

$$WNG = \frac{1}{w^H w}$$

Directivity Index (DI) – Another performance metric that is used for beamformer comparison is the directivity index. The DI basically determines how much power is being radiated in a particular direction with respect to the frequency. For the basis of this report the directionality that will be exclusively considered will be the 2D plane (azimuth only). This is shown in the equation below [15]:

$$DI = 10\log_{10}\left(\frac{|y(\theta_0)|^2}{\frac{1}{2\pi}\int_0^{2\pi}|y(\theta)|^2 d\theta}\right)$$

Where $y(\theta_0)$ is the power from the beamformer directed at a particular azimuth (target) and $y(\theta)$ is the power with respect to the direction (azimuth) [15].

Beam Pattern – To produce the beam pattern for the various frequencies, a fixed reference steering vector of the target was dot product with a scan steering vector across 360 degrees.

$$Y_{magnitude} = conj(s_{target}) \cdot s_{scan}$$

Results

Before testing real data, a simulation of multiple targets is performed to test adaptive beamforming capability. The simulation consisted of two closely spaced signals where target one was a single 250 Hz generated tone at 60 degrees azimuth whereas the 2nd target had broadband with also a single tone at 300 Hz at 10 degrees azimuth. Figure 5 shows the azimuth angle corresponding to each target whilst assuming elevation is 0 degrees.

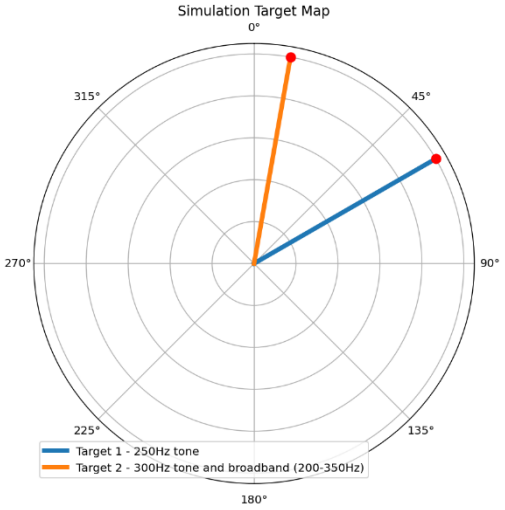


Figure 6. Simulation Target Map

From the two targets a standard single hydrophone cannot distinguish the directivity alone which can be seen in figure 6 where the SNR is lowest among all methods. By utilising 8 elements of the artificially generated hydrophones, the beamformer can suppress the background, hence increasing the SNR as illustrated here. It can be seen that since the beam steers on target one, not only the

background is suppressed but also both beamforming methods attempt to reduce interference (target two) with MVDR having superior results in this example.

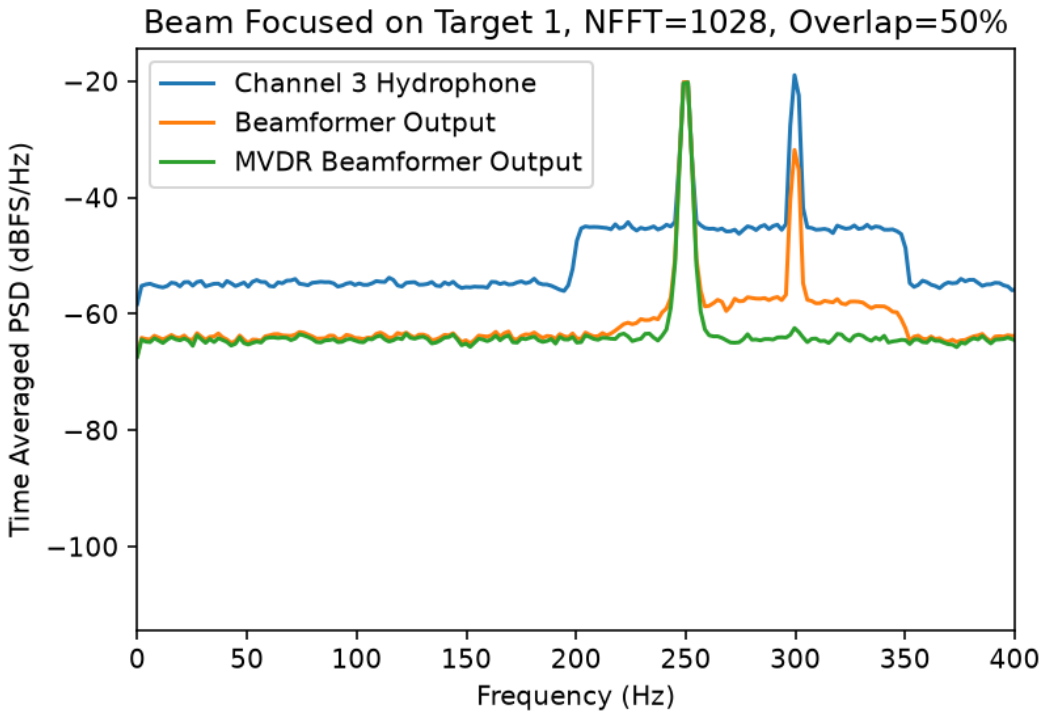


Figure 7. Welch's method graph comparison between all methods for artificial data

The suppression of the interference demonstrates how unwanted noise can be removed using beamformer techniques regardless on the type of signal whether it be a tone or broadband. Figure 8 illustrates a visual representation of a stationary target tone remaining unchanged over time with various techniques mentioned.

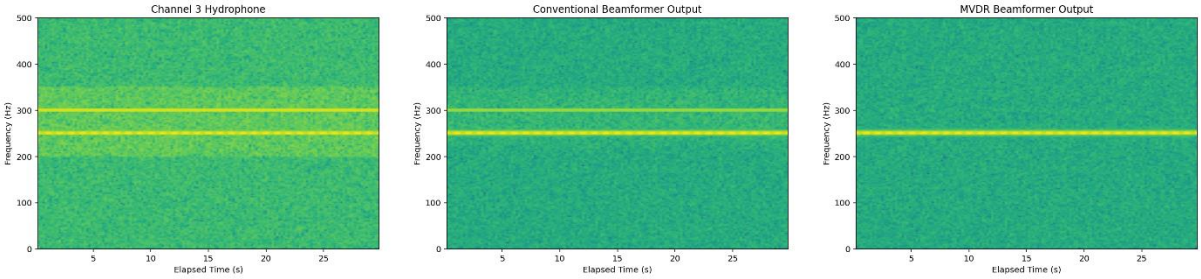


Figure 8. Spectrogram display comparison between all methods for artificial data

To match a real dataset the case of a moving target is considered. A second simulation test is performed by moving target one. The target is moved from its original position at 60 degrees azimuth to 90 degrees as displayed on the simulation moving target in Figure 9. Each sample of the recording the target azimuth changed.

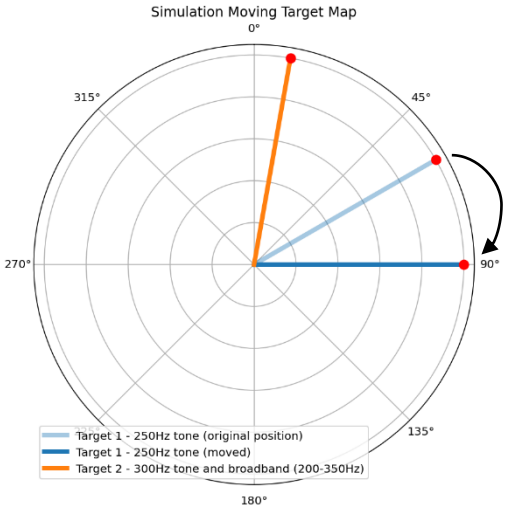


Figure 9. Simulation Moving Target Map

For this analysis a time varying unit vector was required in order keep the beam steered at the target. The results were displayed below.

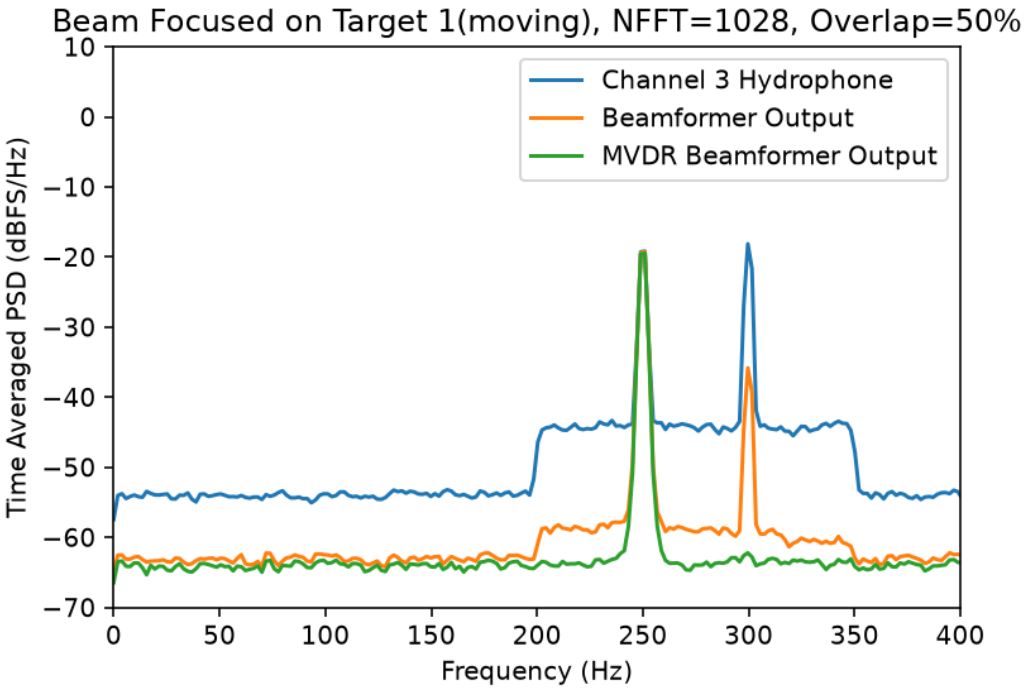


Figure 10. Welch's method graph comparison between all methods for artificial data for moving target

For code validation a package from ARL Python Tools is used to validate and compare both CBF and MVDR beamformer produced results using the same synthetic dataset [17]. For the snapshots required for the covariance matrix in MVDR, the simulation tested various window length to compare performance as well as adjusting the adaptive loading to regularise the matrix.

Table 1. Validation Results

Beamformer	Mean Absolute Error (dB)	Window Length	Adaptive Loading
CBF	1.47e-15	N/A	N/A
MVDR	0.77	81	1
MVDR	0.91	81	0.1
MVDR	0.92	81	1e-3
MVDR	1.47	21	0.1
MVDR	1.11	41	0.1
MVDR	0.97	61	0.1

For one selection of time snapshots for a frequency bin the quality of inverse covariance matrix can be compared.

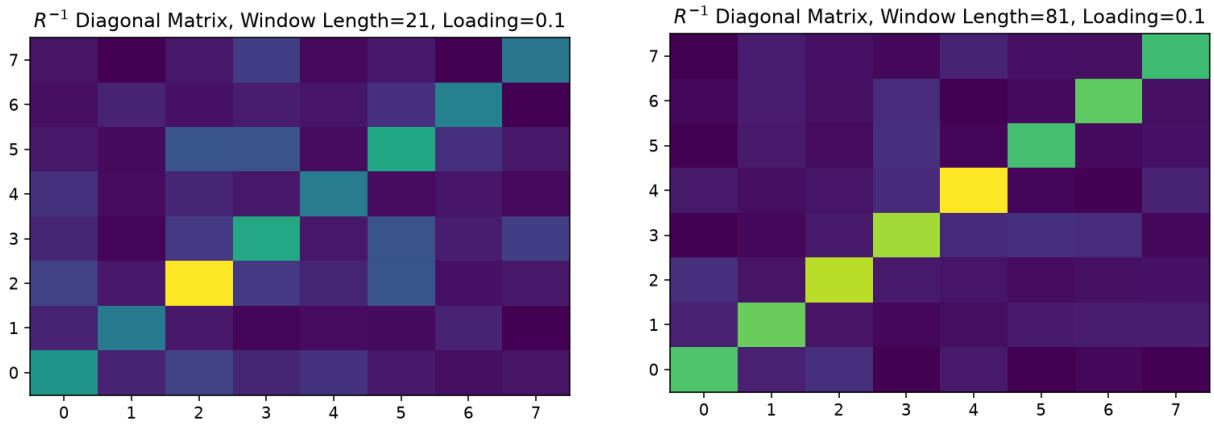


Figure 11. Inverse Covariance Matrix Comparison

The real dataset was analysed using SWellEx-96 experiment which comprised of 4 arrays, one of interest was the Horizontal Linear Array (HLA) for north of the exercise for event S59. The experiment consisted of 32 elements, with 27 working and an aperture of 240m long which was deployed from the north to south with the y-axis mainly used [16]. However, due to the deployment of the array, the “horizontal linear array” had variation in the x-axis and potentially can be considered a random linear array with a fixed depth of 213 meters [16].

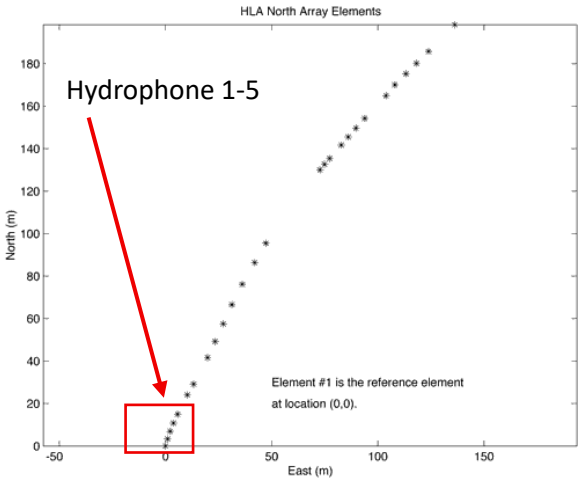


Figure 12. Real data array layout [16]

A map can be seen in figure 13 which highlights the arrays and includes a target noise source and interfering source [16].

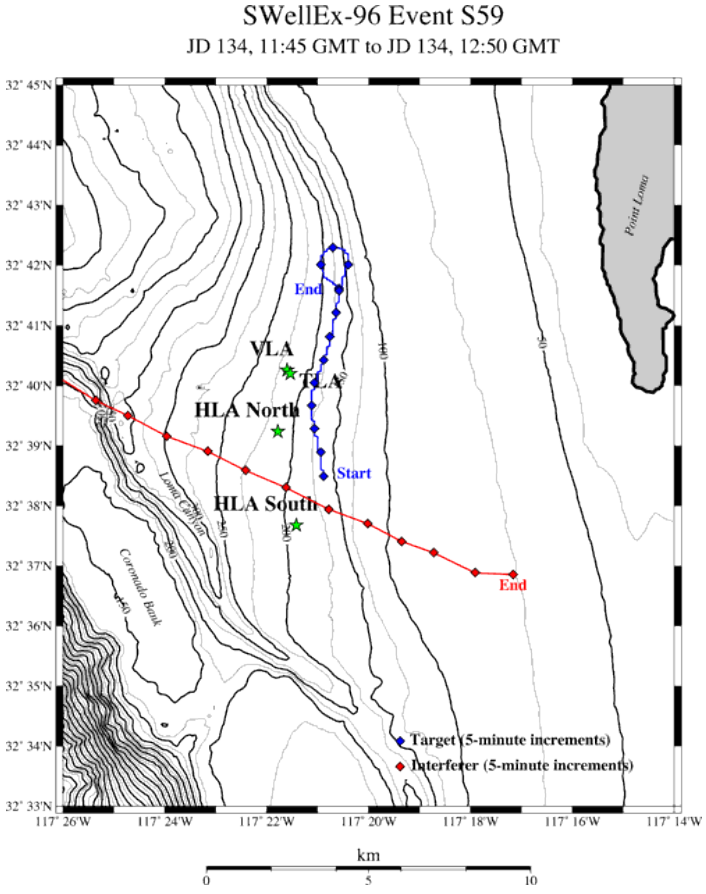


Figure 13. Map overview [16]

A MATLAB script is run to read the "sio" file format where all 27 hydrophones are selected with 100,000 samples which is exported into a mat file. This translates into approximately 30 seconds recording. For the main analysis 5 hydrophones will be used as the focus, with 27 providing a bigger picture in some of the results which will be addressed in the next section. In addition, the acquisition system of the HLA North is listed with a sampling frequency for recording at 3276.8 Hz [16].

One of the issues with the real dataset is target positioning information. Given that the targets such as R/V Sproul are travelling over long distances (multiple kilometres) from the array and the array has a depth of 213 metres, the elevation can be assumed to be a small angle approximately at 0 leaving the azimuth to be scanned. Another issue that the spacing recorded for each hydrophone in the exercise was found to be inconsistent, especially with the select few hydrophones not being processed therefore reducing the potential frequency limits of the system without aliasing. For the basis of the analysis 0 to 400 Hz will still be analysed. However, note that the largest spacing was observed to be approximately 42.92m between elements 15 and 17 which would result into the maximum frequency of 17.5 Hz to avoid spatial aliasing. To minimise this problem a new combination with fewer hydrophones is selected to have the lowest spacing as shown.

Table 2. Real Data Hydrophone Selection [16]

Element	X (m)	Y (m)	Z (m)
1	0	0	0
2	0.95	3.33	0
3	2.24	6.87	0
4	3.78	10.78	0
5	5.68	14.95	0

For the beamforming of real data, due to the position error of the tracking information a frequency of interest is selected after observing best spectrum peaks to beamform from the spectrogram. To find the best beamforming results an azimuth scan search is made across a selected frequency with multiple windows for long recordings to determine the best angles to use for a moving target and one for a short recording for a slow-moving target assuming a fixed azimuth (stationary). For a short recording this is demonstrated below in the following figures. The figure compares the difference between using the fewer elements (5 hydrophones) in a near uniform linear array with all 27 hydrophones in a random linear array.

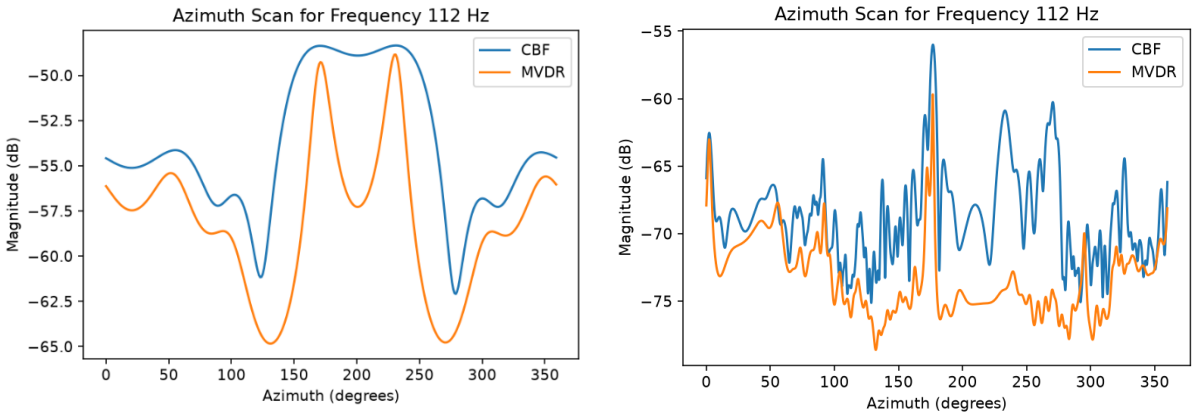


Figure 14. Azimuth scan comparison between hydrophone selection

For various targets, collections of frequencies were collated into the table with the corresponding azimuth. These frequencies were cross checked with the SWellEx-96 experiment results for event S59 [16].

Table 3. Short recording frequency azimuth angle relative to hydrophone axis (elements 1 to 5) [16]

Source	Frequency (Hz)	Azimuth (degrees)
Main	[64, 79, 94, 112, 130, 148, 166, 201, 235, 283, 338, 388]	171
Interference	[62, 77, 92]	294
Other	25	309

Due to computation speed for the moving target a single frequency with individual azimuth was collated into the table. An azimuth scan every 5 seconds was produced for a frequency of 112 Hz and then time averaged with the following peaks angles produced below.

Table 4. Long Recording Track for 112 Hz

Time Elapsed (s)	Azimuth (degrees)
5	173
10	171.2
15	170.3
20	170.8
25	170.9
30	171.7

The main results of the analysis looked at the stationary case across 30 seconds due to the lack of movement observed in the previous section. For the analysis the main and other source were investigated. The first being the other case at 309 degrees azimuth and elevation 0 degrees as

denoted in the figure. The next was the main source at 171 degrees azimuth and elevation also at 0 degrees.

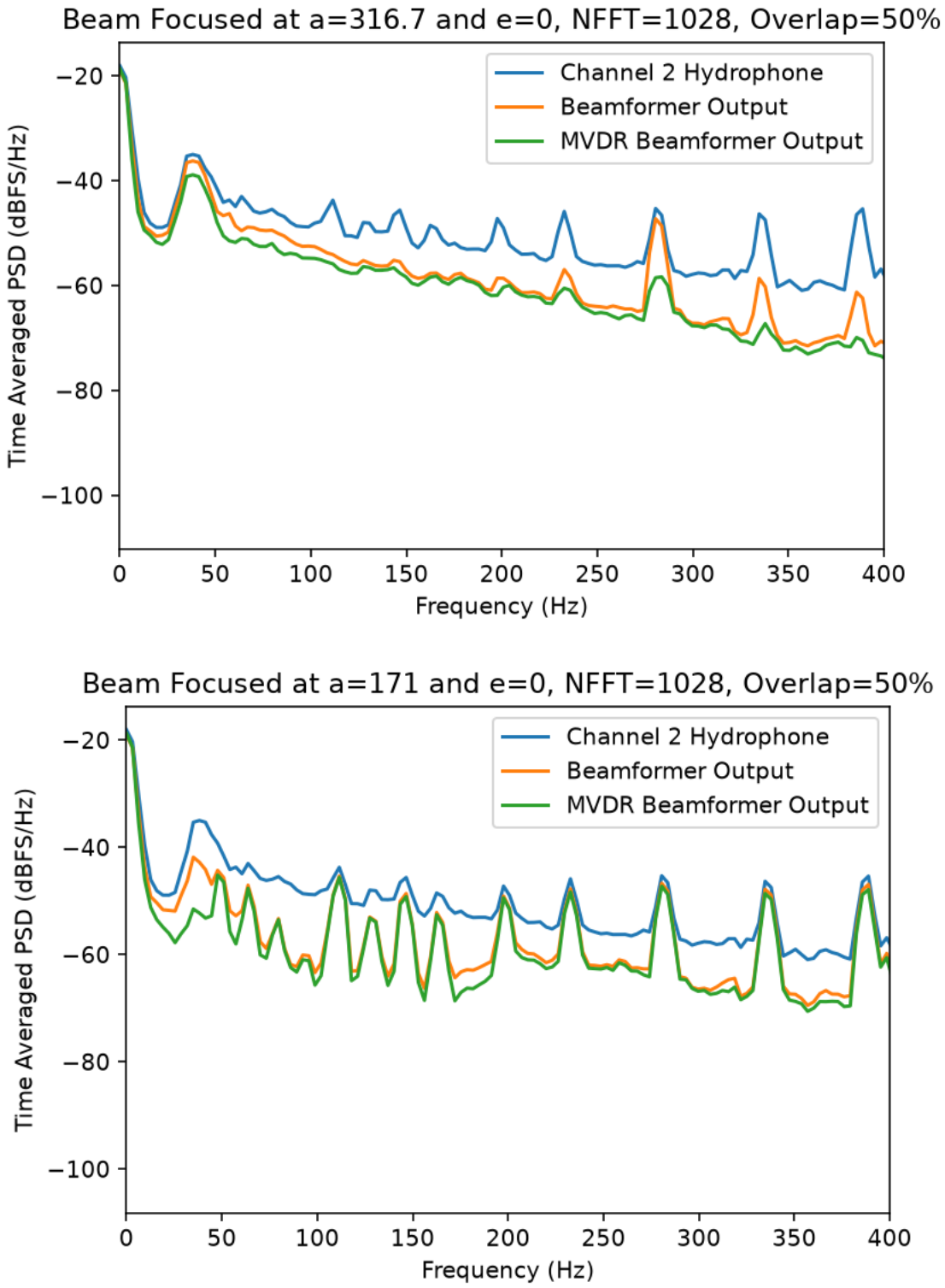


Figure 15. Real dataset comparison across two targets

In addition, spectrograms were produced for the other source and main source respectively.

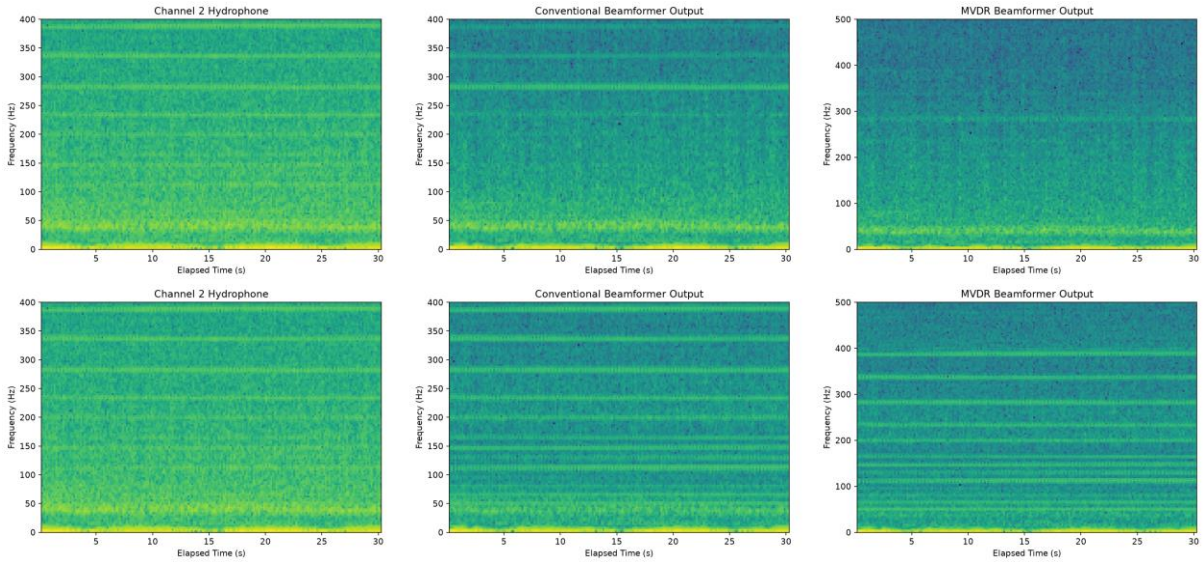


Figure 16. Spectrogram comparison for real dataset

To measure the performance of the beamformer two important performance metrics are considered. The WNG and DI for both conventional beamformer and MVDR. For this performance analysis, the main source will be considered.

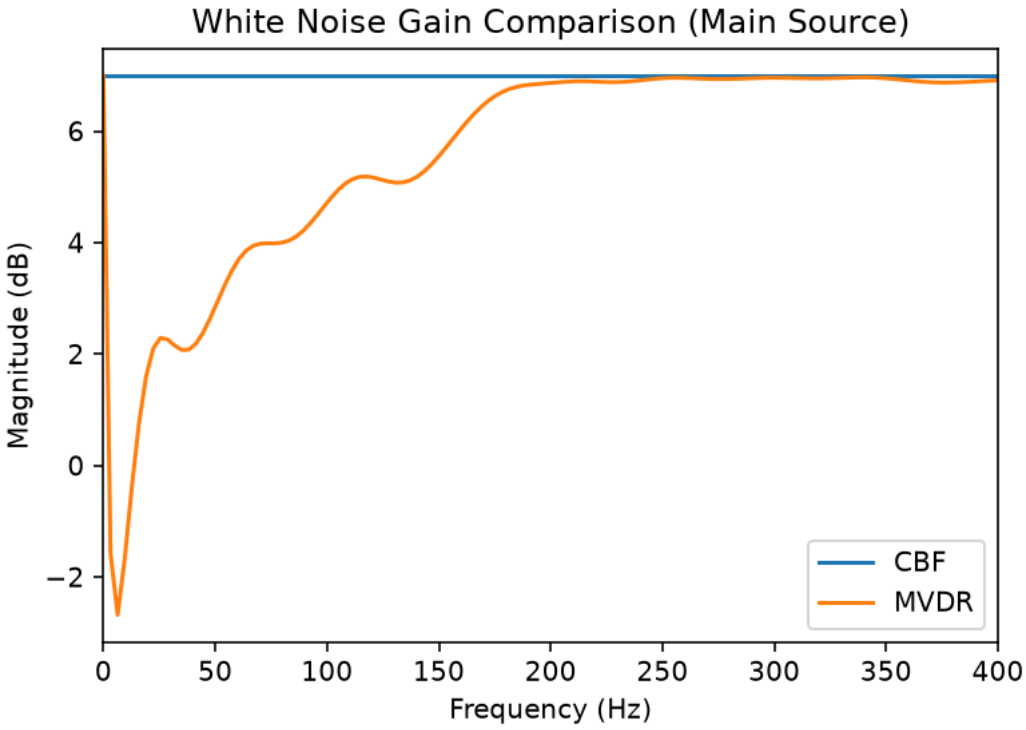


Figure 17. White noise gain comparison

The DI was performed with the average azimuth scan across 0 to 360 degrees for the nearest degrees. The analysis was performed for the main source which was 171 degrees azimuth and 0 degrees elevation.

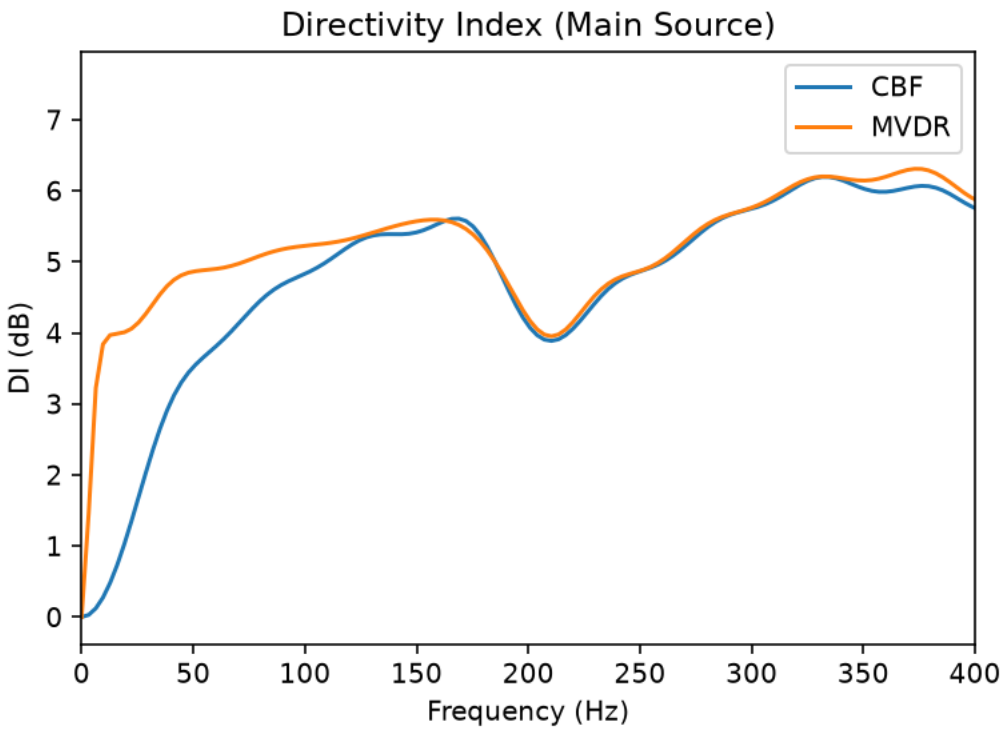


Figure 18. 2D directivity index comparison

A generalised beamformer pattern was produced with respect to 10 Hz, 100 Hz, 200 Hz and 300 Hz at an elevation and azimuth of 0 degrees. For the MVDR beamformer, the weights were computed using the diffused correlation matrix.

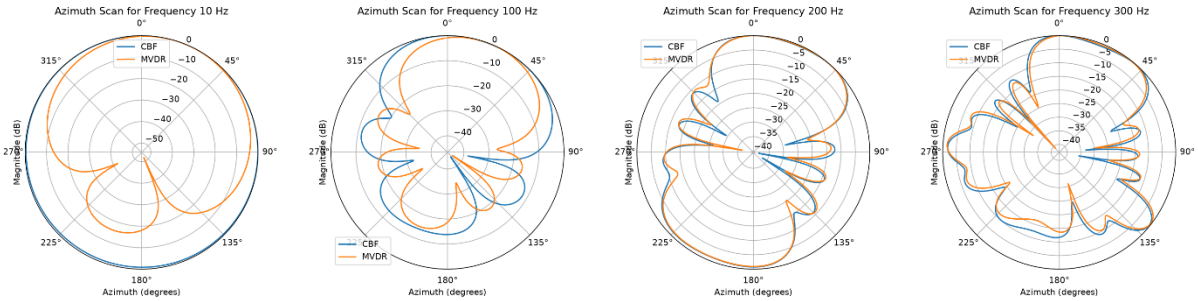


Figure 19. Generalised beampattern comparison

Discussion

The simulation demonstrated how sources in a spectrum injected with noise which includes white noise and other interference were able to be suppressed. This can be seen in the figure that with the beamformer directed at target one with the noise floor drastically reduced from approximately -54 dB to -63 dB as illustrated whilst preserving the 250 Hz main tone hence showing a signal to noise increase. This coincides with the theoretical expected array gain (AG) of 9 dB from using 8 elements in a uniform linear array. In addition, the 300 Hz noise tone has been slightly reduced by the conventional beamformer and been completely removed for the MVDR. The conventional beamformer having uniform weights although the white noise gain has been suppressed, the interference source was present in the spectrum. This is caused by the beamformer pattern potentially including the nearby target in the main lobe as well as the side lobes. In figure 8 the spectrogram display showed each signal source remained consistent over time as expected for the

stationary case. The beamforming methods show the broadband segment of energy and tones being completely removed in the MVDR display.

The next simulation involving the moving target also produced similar results with very slight variation. The conventional beamformer in the moving target scenario showed more difficulty in dealing with the broadband signal as opposed to when the target was stationary. In addition, upon zooming into the MVDR peak it was found that the 250 Hz peak did not entirely match the signal of the hydrophone exactly with a slight difference of approximately 0.3 dB. This small error shows how sensitive the MVDR is to the positioning information for the steering vector. For this example, because the simulated moving target was produced at intervals of the whole recording, not interpolated at each time bin there is a small error introduced which can be seen.

The validation of the algorithm implementation generated a series of interesting results. For the conventional beamformer since no covariance matrix is required additional window length or adaptive loading is not needed. When compared against ARL Python Tools, an absolute mean error of 1.45e-15 dB was found having the best results. However, for the MVDR beamformer variable results were produced based which were heavily influenced based on the composition of the covariance matrix. As a greater window length was used the mean absolute error decreased from 1.47 dB to 0.91 dB with a fixed adaptive loading of 0.1. Figure 11 illustrates how the inverse covariance matrix can be observed to be ill conditioned when using a smaller window length as opposed to a larger one where the diagonal components are more pronounced than spread randomly. Nevertheless, the comparison was made for a slow-moving source. However, for a very fast target this may not be the case as the covariance matrix may be overestimated using the larger localised time frames along the moving window and hence capture the absence of the target.

Similarly, at a fixed window length of 81 it was observed that adjusting for a larger adaptive loading to the covariance matrix improved the error of the results. By increasing the adaptive loading from 1e-3 to 1 it was seen that the absolute mean error dropped from 0.92 dB to 0.77 dB. Although a lower absolute error was achievable, the MVDR begun to behave like the conventional beamformer losing the value of nulling the interference.

Originally for the real data all 27 hydrophones were utilised for the analysis. However, due to the smaller inconsistent spacing led to potential aliasing of results for higher frequencies. Subsequently, the main analysis consisted with 5 selected hydrophones in event S59, it was confirmed a series of tonals were identified throughout the track. Both the main source target and interference were both present with an additional other source.

The main target was estimated to have a peak azimuth at approximately 171 degrees to the nearest degree through an azimuth scan this coincided with the map overview in figure 13. This was determined by comparing the azimuth angles with each reported main source tone. For this example, a tone of 112 Hz is selected. On the left side of figure 14 it can be observed that there were two observed azimuth peaks with the fewer hydrophone setup with angles 171 and 230 degrees respectively due to the linear array ambiguity. This was overcome by utilising the characteristics of the 27 random linear array elements to resolve the correct angle.

Upon comparing the frequency of the series of known tones reported for the main source it was found that the 49 Hz was unable to be identified clearly. This could be due to a few factors such as lower angular resolution from the aperture of the array low signal level. In addition, the interference source failed to give consistent azimuth peaks from 107 Hz and above. This could be explained by

also the very low signal level being overwhelmed by the main source as well as spatial aliasing at the higher frequency end.

For a moving target there was similar results observed. For the main source there was a decrease in azimuth from 173 to 170.3 degrees from 5 to 15 seconds then an increase to 171.7 degrees after 30 seconds. These angles were observed to not change as much relative to the stationary case from the main source found earlier which could be explained by the short duration of the recording of the data as well as the distance of the target resulting into a small angle change.

The main analysis results showed consistency in both the main and other source. It can be seen that when the beam is directed at the 25 Hz source the spectrum tries to remain flat in the MVDR beamformer, however, struggles with the tones near 300 Hz and 350 Hz respectively. This potentially could be improved with longer window lengths and appropriate adaptive loading to the covariance matrix as addressed in the simulation results. In addition, the conventional beamformer lacks the adaptive null weights needed to reduce the interfering signals. For the main source it can be seen that the beamformer performance in both MVDR and conventional beamformer displays high level of unwanted noise being reduced. In addition, the tone at 25 Hz of the other source has been clearly suppressed as shown. From this reduction in noise, it can be seen in figure 16 that the spectrogram is now displaying increased detail of information across the frequency band in relation to the target that the beam is steered directly to.

To emphasise the improvements in the removing unwanted noise, performance metrics were produced for WNG and DI which showed definite improvements in the SNR with slight degradation in performance in other areas. In figure 17 it can be seen that the main source conventional beamformer remained consistent across all frequencies with a WNG gain of 7 dB whereas for the MVDR beamformer dropped at the lower frequency band to a minimum of approximately -2.7 dB, however, subsequently remained consistent. As a result, this negative gain will lead to noise amplification to the affected frequency band. This lower frequency performance can be explained by the covariance matrix attempting the differentiate the very similar phase differences making it hard to correlate the signals.

The 2D plane DI performance had an inverse effect in the results. It can be seen in figure 18 that although the conventional beamformer follows a similar shape to the MVDR beamformer, the DI is significantly increased by 4 dB initially with slight improvements across all the frequency bands. This increase of the DI can be explained by the trade off with the WNG found previously, illustrating how spatial information can be used differently.

The beam patterns were also analysed for both conventional and MVDR beamformers. It can be seen that at the best-case scenario the MVDR performance shape still retains strong main lobe shape at low frequencies (10 Hz), however, progressively become similar to a conventional beamformer at 300 Hz. The peak performance of the MVDR beamformer can be seen at 100 Hz by the narrowest beamwidth and smaller side lobes.

Note that given the ideal conditions of the MVDR algorithm following a superdirective beamformer, a beam scan on known targets with nearby interference will result into different beam pattern shapes. This includes the main lobe beamwidth broadening. It can also be seen when the beamformer is directed at 0 degrees azimuth, the beam orientation at 0 degrees is shifted clockwise slightly towards the endfire, this is due to the array geometry of the x-axis.

Conclusion

Through each of the analysis in comparing the performance of the MVDR with respect to the conventional beamformer it was found that MVDR provided excellent results in identifying the directivity of noise sources which was evident in the azimuth scan for particular frequencies as well as in the performance metric results which showed an improvement in DI for the lower frequency compared to the conventional beamformer. Although WNG was slightly degraded, the benefits showed substantial difference in reducing other unwanted noise from underwater recordings.

Furthermore, MVDR covariance matrix regularisation was briefly explored. For optimum results it was found that selecting the larger window lengths for the covariance matrix as well as adaptive loading improved the results, however, increasing both parameters too high lead to degradation in performance.

Ultimately, both beamformers could achieve the goal of suppressing noise, however, with real underwater acoustic data MVDR can be limited if the target is not well understood. Slight variation in the steering vector error could lead to poor sensor correlation in the covariance matrix compromising the weights and potentially adding an unwanted null. Hence, highlighting one of the challenges of working with adaptive beamforming. Nevertheless, with accurate positioning information and close monitoring of regularisation of the covariance matrix it can be seen that MVDR still remains to be one of the most essential algorithms to use with underwater signals.

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